**CSA5108 Cryptography and Network security**

**Name :** K. Sirisha
**Reg.no:**192325130

**Exp 17 :** DES Decryption — Reverse Round-Key Order
**Code:**

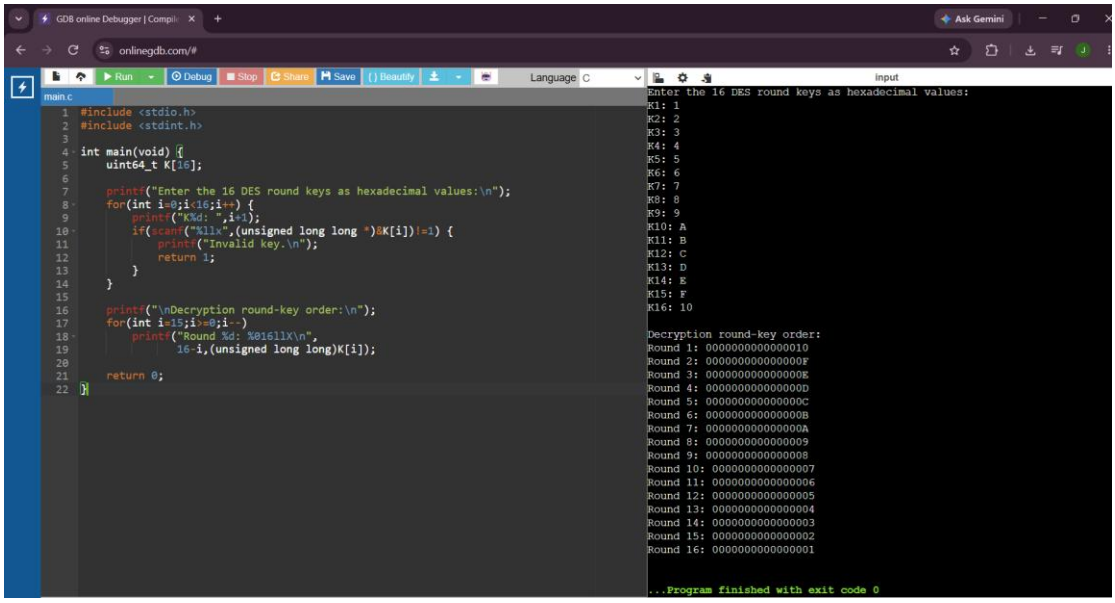
```
#include <stdio.h>
#include <stdint.h>

int main(void) {
   uint64_t K[16];

   printf("Enter the 16 DES round keys as hexadecimal values:\n");
   for(int i=0;i<16;i++) {
      printf("K%d: ",i+1);
      if(scanf("%llx",(unsigned long long *)&K[i])!=1) {
         printf("Invalid key.\n");
         return 1;
      }
   }

   printf("\nDecryption round-key order:\n");
   for(int i=15;i>=0;i--)
      printf("Round %d: %016llX\n",
           16-i,(unsigned long long)K[i]);

   return 0;
}
```


GDB online Debugger | Compil
Ask Gemini
onlinegdb.com/#
Run
Debug
Stop
Share
Save
{ } Beautify
Language C
main.c
#include <stdio.h>
#include <stdint.h>
int main(void) {
uint64_t K[16];
printf("Enter the 16 DES round keys as hexadecimal values:\n");
for(int i=0;i<16;i++) {
printf("K%d: ",i+1);
if(scanf("%llx",(unsigned long long *)&K[i])!=1) {
printf("Invalid key.\n");
return 1;
}
}
printf("\nDecryption round-key order:\n");
for(int i=15;i>=0;i--)
printf("Round %d: %016llX\n",
16-i,(unsigned long long)K[i]);
return 0;
}
input
Enter the 16 DES round keys as hexadecimal values:
K1: 1
K2: 2
K3: 3
K4: 4
K5: 5
K6: 6
K7: 7
K8: 8
K9: 9
K10: A
K11: B
K12: C
K13: D
K14: E
K15: F
K16: 10
Decryption round-key order:
Round 1: 0000000000000010
Round 2: 000000000000000F
Round 3: 000000000000000E
Round 4: 000000000000000D
Round 5: 000000000000000C
Round 6: 000000000000000B
Round 7: 000000000000000A
Round 8: 0000000000000009
Round 9: 0000000000000008
Round 10: 0000000000000007
Round 11: 0000000000000006
Round 12: 0000000000000005
Round 13: 0000000000000004
Round 14: 0000000000000003
Round 15: 0000000000000002
Round 16: 0000000000000001
...Program finished with exit code 0